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ARCHITECTURE JUSTIFICATION

Architecture Selection — Municipal Solid Waste Route Optimizer

We chose Layered Architecture for the Municipal Solid Waste Route Optimizer.

Reason 1

The system has clearly separated responsibilities such as bin monitoring, route optimization, fleet management, complaint management, and data management. Layered architecture provides separation of concerns, making the system easier to understand, maintain, and modify.

Reason 2

The architecture allows the route optimization functionality to be separated from data management and user-facing operations. This makes the system suitable for handling information from smart bins while independently managing collection routes and truck operations.

SECURITY ADVANTAGE

Security Advantage: Layered architecture separates application processing from the data management layer. This allows access to sensitive operational data to be controlled through the appropriate application components rather than allowing direct access by users.

PERFORMANCE ADVANTAGE

Performance Advantage: The route optimization functionality is separated into its own component, allowing route calculations to be processed independently from complaint management and other system functions. This supports efficient route recalculation and helps the system meet the requirement of recalculating routes for 50 trucks and 10,000 bins within 45 seconds.